

AthleteIQ

Founder's Launch Playbook

Pre-Launch Checklist + Worst-Case Disaster Scenarios

What to fix before going live with 50–100+ athletes

26

items to check

12

disaster scenarios

3

priority tiers

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"Launching is 90% firefighting. This document is your fireproof shield."

CRITICAL Complete before a single real user touches the app

Set WEB_CONCURRENCY=1 on Render **RAM**

- ☐ Add in Render env vars. Multiple workers fight for 512MB RAM and crash the service under concurrent load.

12-hour AI insight caching **RAM** **Groq cost**

- ☐ Check Supabase before calling Groq. If a row for this athlete exists within 12 hrs, return it. Single biggest RAM + cost saver.

Set max_tokens=250 on every Groq call **Groq cost**

- ☐ Uncapped Groq calls can return 1000+ tokens. 250 is enough for a clear narrative and costs 4x less.

Make all Supabase calls async (await) **RAM**

- ☐ A synchronous DB call blocks the single Uvicorn worker. Every other request queues up and times out.

Switch FastAPI log level to WARNING **RAM**

- ☐ Logging every 200 OK floods RAM and disk I/O on free Render. Set uvicorn log_level='warning' in startup command.

Pre-calculate ACWR in FastAPI before Groq **Groq cost**

- ☐ Never send raw check-in rows to Groq. Compute ACWR, readiness score, risk tier in Python — send only the 3-line result to narrate.

HIGH Complete before your first paying coach signs up

trial_ends_at auto-populated on every signup **Security**

- ☐ Every new academy row must get trial_ends_at = created_at + 14 days. Verify in Supabase on an existing test academy row.

Soft gate on bulk_log and drill_centre endpoints **Security**

- ☐ Add a FastAPI dependency that checks plan_status and trial_ends_at. Return 403 with a clear message if expired.

Frontend cooldown on Refresh Advice button **Groq cost**

- ☐ Track last_fetched in localStorage. If <30 min, show 'Updated X mins ago'. Protects Groq quota from impatient parents.

Soft deletes — add is_deleted to athletes + coaches **Security**

- ☐ Never hard DELETE. Set is_deleted=true and filter in queries. A coach wiping their squad before a match is a disaster.

Lock CORS to your Vercel domain only **Security**

- ☐ Remove the wildcard * from FastAPI CORS. Set allow_origins to your exact Vercel URL. Wildcard = anyone can call your backend.

Add Supabase RLS (Row Level Security) policies **Security**

- ☐ Without RLS, a user can bypass FastAPI and query any academy's data via the Supabase JS client directly.

Add UptimeRobot ping every 14 minutes **RAM**

- ☐ Free Render spins down after 15 mins. A free UptimeRobot monitor on /health keeps it warm. Without this, first daily request takes 40+ sec.

MEDIUM Complete before scaling beyond 30 athletes per day

Razorpay webhook: verify HMAC-SHA256 signature **Security**

- ☐ Anyone can POST fake payment data to your webhook. Validate the Razorpay-Signature header. Without this, users can fake a payment.

Supabase connection pooling (PgBouncer) RAM

- ☐ Free tier limits direct connections. At 50 concurrent users you hit the cap. Enable the pooler URL in Supabase and use it in FastAPI.

6-digit access code for parent login UX

- ☐ Replace child-name login with a unique 6-digit code per athlete. Generate on creation, store in DB. More private and professional.

Add Sentry free tier for error tracking UX

- ☐ Right now if your backend crashes silently you find out when a coach WhatsApps you. Sentry catches every exception and emails you instantly.

Audit all Log/Submit buttons — minimum 44px height UX

- ☐ Open Chrome DevTools in mobile mode. Check every primary action button. Under 44px = a coach with sweaty hands will miss it mid-session.

High-contrast injury risk alerts UX

- ☐ Injury Risk badges must pass WCAG AA contrast in direct sunlight. Red-on-white or amber-on-dark, font-weight 600. Test it outside.

WhatsApp support button in dashboard footer UX

- ☐ Add a 'Need help?' button opening wa.me/your_number. In early sales, personal support is your biggest edge over any bigger app.

Input sanitisation on all POST endpoints Security

- ☐ Strip HTML/script from coach_notes, athlete_name, and free-text fields before Supabase. A curious athlete could inject malformed data.

Add /health endpoint to backend RAM

- ☐ GET /health returning {status: ok} is needed for UptimeRobot, Render health checks, and future monitoring. Takes 2 minutes.

Create Supabase backup schedule Security

- ☐ Enable point-in-time recovery or set a weekly pg_dump export. If a coach's data is corrupted, you need a restore point.

What Could Go Wrong

Worst-case scenarios for a live launch with 50–100+ athletes

Read this before you go live. Every scenario below has happened to a real startup.

CRITICAL

HIGH

MEDIUM

#01The Render Death SpiralInfrastructureCRITICAL

SCENARIO At 7 AM, three coaches across Pune hit Bulk Log simultaneously for their morning squads — 50+ athletes total hitting your API in a 60-second window.

RESULT Render's 512MB RAM limit is breached. The server hard-restarts mid-request. Coaches see '502 Bad Gateway'. They assume the app is broken and go back to WhatsApp and paper.

SOLUTION `SERVERS_CONCURRENCY=1`. Add a `/health` endpoint. Set up UptimeRobot. Consider Render Starter (\$7/mo) the moment first Pro coach pays.

Likelihood: Very High

Impact: Reputation-killing on Day 1

#02The Groq API BankruptcyAI EngineCRITICAL

SCENARIO Parents get 'unlimited AI insights'. Parents get excited and refresh the Recovery Advice page every 5 minutes. 10 parents x 12 refreshes/hr = 120 Groq calls/hr from one academy alone.

RISK Rate-limits your API key. All AI features — ACWR, Drill Centre, Parent Portal — stop working for every academy simultaneously. No warning, no fallback.

SOLUTION Your own server-side cache. Frontend cooldown button. `max_tokens=250`. ACWR pre-calculation in FastAPI before touching Groq.

Likelihood: High

Impact: All AI features go dark

#03The 40-Second Cold StartInfrastructureCRITICAL

SCENARIO Early morning between 11 PM and 6 AM. Render spins the service down. A coach arrives at the ground at 6:15 AM, opens the app to check readiness scores before training.

RESULT The app hangs for 35-45 seconds while Render boots. The coach thinks the app has crashed. They tell every parent at that session the app doesn't work.

SOLUTION UptimeRobot free monitor pinging `/health` every 14 minutes. This keeps the server warm 24/7 at zero cost.

Likelihood: Certain without fix

Impact: First impression destroyed

#04The Deception Flag Miss & Liability NightmareSafety & LegalHIGH

SCENARIO A parent wants to play in a district tournament but has a hamstring strain. They report 10/10 readiness on every metric to hide it. AthleteIQ gives them a green light. They play, the muscle tears completely.

RESULT The parent blames AthleteIQ's 'AI clearance'. Legal threats. Your Shiv Chhatrapati reputation takes a hit in a community where word travels in hours.

SOLUTION Medical disclaimer on every readiness score: 'This is a workload tool, not a medical diagnosis.' Never use the word Cleared. Use 'Appears Ready'.

Likelihood: Medium

Impact: Legal threat + reputation loss

#05The Parent Data LeakSecurityHIGH

SCENARIO Athletes have sequential IDs (athlete/101, athlete/102). A tech-savvy parent guesses the URL pattern and views another child's mental health notes, injury flags, and recovery advice.

RESULT In a tight-knit Pune academy, this spreads in one WhatsApp group. You lose the entire academy — all coaches, all parents — in under 24 hours.

SOLUTION Your own IDs for all athlete IDs. 6-digit access codes for parents. Supabase RLS policies so the database itself enforces isolation.

Likelihood: High without fix

Impact: Entire academy lost overnight

#06

The Supabase 'Too Many Clients' Crash

Database

HIGH

SCENARIO 10+ concurrent users — coaches logging in, athletes checking in, parents viewing portals — all creating direct DB connections. Supabase free tier has a hard connection limit.

RESULT Requests get 'sorry, too many clients already' errors. The dashboard loads blank. Athletes see empty data. Coaches panic that all their records are gone.

ENABLE PgBouncer connection pooler in Supabase project settings. Use the pooler connection string in FastAPI instead of the direct string.

Likelihood: Certain at 50+ users Impact: App appears broken for everyone

#07

The Sweaty Phone Data Loss

UX / Reliability

HIGH

SCENARIO Coach at a rink with spotty 4G. They complete the 3-step Bulk Log for 25 athletes. The POST request hangs for 8 seconds then silently times out. The coach closes the app.

RESULT Server never received the data. When the coach opens the dashboard later, the session is missing. They spent 3 minutes logging for nothing. Trust gone.

ADD Loading state that clearly says 'Saving...' and an explicit 'Saved!' confirmation. Show an error toast on failure with a retry button. Never let the user guess.

Likelihood: High in Indian 4G conditions Impact: Core feature feels broken

#08

The Trial Cliff on Match Morning

Business / UX

HIGH

SCENARIO 14-day trial ends at midnight on a Sunday. Monday 5:30 AM — match morning — they open the app to log the warm-up session and hit a hard paywall.

RESULT They cannot log the session. They have no card handy. They are stressed and angry. They uninstall. You never hear from them again.

ENABLE Grace period after trial ends — features stay active but show a persistent banner. Send WhatsApp/email reminders on days 10, 12, 13. Soft block on day 17.

Likelihood: High Impact: Churn at the worst moment

#09

The Fake Payment Attack

Security / Revenue

MEDIUM

SCENARIO Competitor discovers your /webhook/razorpay endpoint. They craft a fake payment.captured payload and POST it. Your backend updates the academy plan to Pro without any payment.

RESULT You are providing Pro features — AI engine, Drill Centre, unlimited athletes — for free. At scale, this is revenue theft and a free-rider problem.

VALIDATE Every webhook using HMAC-SHA256 signature with your Razorpay webhook secret. Reject any request where signatures don't match.

Likelihood: Medium Impact: Revenue loss + free riders

#10

The AI Hallucination in a Recovery Plan

AI Quality

MEDIUM

SCENARIO AI generates a recovery advice narrative for a 14-year-old athlete that includes something medically inaccurate — wrong rest duration, incorrect nutrition advice, or contradicts a physio's plan.

RESULT Parent follows AI advice. Athlete's recovery is delayed. Parent complains publicly in the academy WhatsApp group. Coach distances themselves from AthleteIQ.

ADD Visible disclaimer under every AI output: 'AI-generated guidance — always consult your coach and physio.' Frame advice as suggestions, not prescriptions.

Likelihood: Medium Impact: Trust collapse in one message

#11 The CORS-Broken Third-Party Access

Security MEDIUM

SCENARIO wildcard CORS (*) active. A malicious script on a third-party website makes authenticated API calls to your backend using a stolen session token from a coach's browser.

RESULT Site request forgery — the attacker can read athlete data, post fake check-ins, or trigger AI calls at your expense.

ADVICE `allow_origins` in FastAPI to your exact Vercel URL only. Never use * in production.

Likelihood: Medium Impact: Data breach, API abuse

#12 The Silent Backend Crash

Infrastructure MEDIUM

SCENARIO edge case in your ACWR calculation — an athlete with zero training logs — causes an unhandled division-by-zero exception. FastAPI returns 500 errors but you have no alerting.

RESULT Found out when 5 coaches WhatsApp you at the same time. You have no error details, no stack trace, no way to reproduce it quickly. The fix takes hours instead of minutes.

ADVICE Sentry free tier. One line of code — `sentry_sdk.init(dsn='...')` — captures every unhandled exception with full stack trace and emails you within 30 seconds.

Likelihood: Certain eventually Impact: Extended downtime without alerting